CSCI 4623/5623 **DIGITAL FORENSICS**

THE UNIVERSITY *of* NEW ORLEANS

02: DIGITAL FORENSICS

DEFINITIONS & EXAMPLES

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WHAT IS ***FORENSIC SCIENCE***?

aka FORENSICS

Forensic science

Forensic science is the application of scientific methods to collect, preserve, and analyze evidence related to legal cases.

- Historically,
  - it referred to the examination of samples of ***physical*** evidence to establish causal sequence of events;
  - of special interest are the establishment of ***provenance*** and ***authenticity*** of the evidence.
- Rationale: *Locarde's exchange principle*
  - physical contact between objects inevitably results in the exchange of matter leaving traces that can be analyzed to (partially) reconstruct the event

DIGITAL FORENSIC SCIENCE

- Same basic assumption applied to the digital/cyber world
  - i.e., cyber systems are **assumed** to exhibit "exchange" behavior → **not** necessarily true, and analysts must understand how the system functions;
  - still, a lot of useful evidence ***is*** being collected;
  - a lot of it is insecure → rarely challenged in court, but it *could* be manipulated so additional reasoning is required for a sound analysis.

THE *DAUBERT* STANDARD

- Three landmark cases:
  - Daubert v. Merrell Dow Pharmaceuticals, 509 U.S. 579 (1993);
  - General Electric Co. v. Joiner, 522 U.S. 136 (1997);
  - Kumho Tire Co. v. Carmichael, 526 U.S. 137 (1999).
- Superseeds the *Frye* standard
  - Frye v. United States, 293 F. 1013, D.C. Cir. 1923
- Goodstein*:
  - "The presentation of scientific evidence in a court of law is a kind of shotgun marriage between the two disciplines. … The Daubert decision is an attempt (not the first, of course) to regulate that encounter."

*Goodstein, D. Reference Manual on Scientific Evidence: Third Edition. 2015, pp. 37-54.
https://www.fjc.gov/sites/default/files/2015/SciMan3D01.pdf

DAUBERT CRITERIA

1. The theoretical underpinnings of the methods must yield testable predictions by means of which the theory could be falsified.
2. The methods should preferably be published in a peer-reviewed journal.
3. There should be a known rate of error that can be used in evaluating the results.
4. The methods should be generally accepted within the relevant scientific community.

NB: Trial judges have ***a lot*** of leeway in deciding what evidence is admitted.

Digital Forensics definitions

- DFRWS (2001)
  - "The use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations."

Digital Forensics definitions

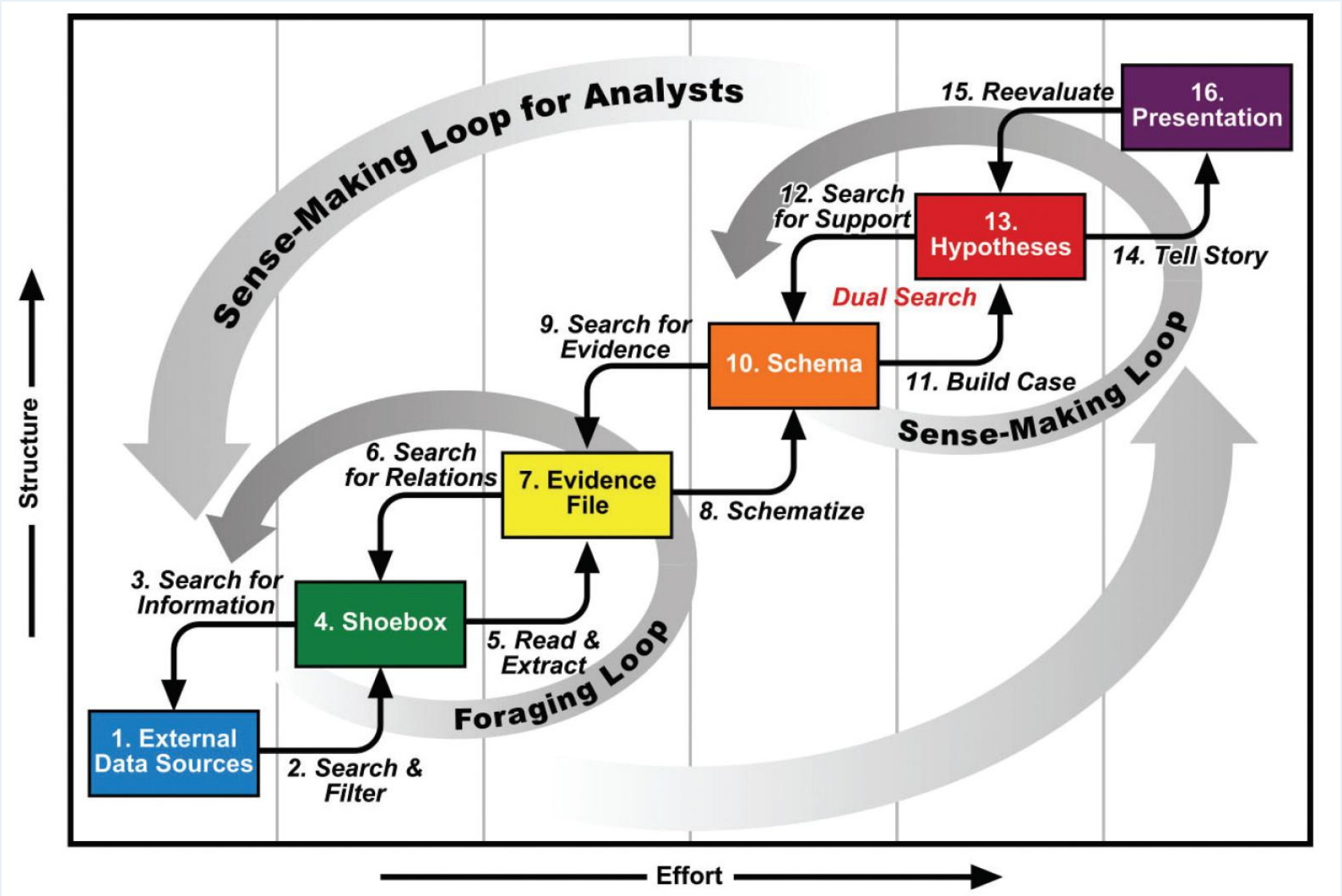
- NIST (2006)
  - "The application of science to the identification, collection, examination, and analysis of data while preserving the integrity of the information and maintaining a strict chain of custody for the data. Data refers to distinct pieces of digital information that have been formatted in a specific way."

WORKING **TECHNICAL** DEFINITION

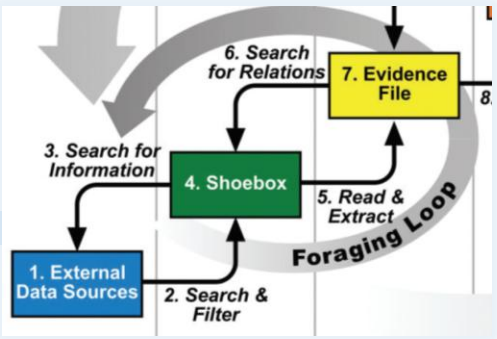
- Roussev (2015)
  - » "The process of reconstructing the relevant sequence of events that have led to the currently observable state of a target IT system or (digital) artifacts."
- Notes:
  1. The notion of relevance is inherently case-specific, and a big part of forensic analysts expertise is the ability to identify case-relevant evidence.
  2. Frequently, a critical component of the forensic analysis is the causal attribution of event sequence to specific human actors of the system (such as users and administrators).
  3. The provenance, reliability, and integrity of the data used as evidence is of primary importance.

I.E.: We view all efforts to perform system, or artifact, ***analysis after the fact*** as a form of forensics. This includes common activities, such as incident response and internal investigations that almost never result in any legal actions

Cognitive task model (Pirolli & Card)



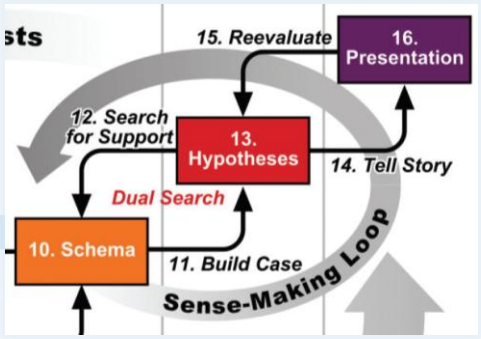
Cognitive task model (2)



- *External data sources* include all potential evidence sources for the specific investigation, such as disk images, memory snapshots, network captures, as well as reference databases, such as hashes of known files.
- The *shoebox* is a subset of all the data that has been identified as potentially relevant, such as all the email communication between two persons of interest.

  At any given time, the contents of the shoebox can be viewed as the analyst's approximation of the information content potentially relevant to the case.
- The *evidence file* contains only the parts that directly speak to the case, such as specific email exchanges on the topic of interest.

Cognitive task model (3)



- The *schema* contains a more organized version of the evidence, such as a timeline of events, or a graph of relationships, which allows higher-level reasoning over the evidence.
- A *hypothesis* is a tentative conclusion that explains the observed evidence in the schema and, by extension, could form the final conclusion.

  Once the analyst is satisfied that the hypothesis is supported by the evidence, the hypothesis turns into a presentation, which is the final product of the process.
- The *presentation* usually takes on the form of an investigators report that both speaks to the high-level conclusions relevant to the legal case, and also documents the low-level technical steps based on which the conclusion has been formed.

CTA: Analytical processes

- Foraging loop
  - actions taken to find potential sources of information, query them, and filter them for relevance;
- Sense-making loop
  - actions taken to analyst develop (in an iterative fashion) a conceptual model that is supported by the evidence.
- Two types of processes
  - bottom-up (synthesis): organizing data to build a theory
  - top-down (analysis): finding data based on a theory & objectives.

  In practice, analysts apply these in an opportunistic fashion with many iterations.

TurboChef vs. LickMySpoon*

A Case Study

*scenario by Golden G. Richard

Case Overview

- Claim:
  - TurboChef's secret recipes are being used by LickMySpoon, LLC
- Suspected crime:
  - Intellectual property theft
- Incident response:
  - 10/26/2004
- Case credit:
  - Golden G. Richard

Case Background

- Daryl Popper
  - » exclusive custodian of secret recipes
  - » uses ***bcrypt*** to encrypt/decrypt recipes
  - » recipes kept encrypted unless being modified
- Cyber environment
  - » single ***administrator*** account as login / **no** password
  - » connected to internet
  - » no network file sharing, because no appropriate accounts set up
  - » spare company laptops available
  - » company USB keys issued for file transfers
- Initial target ➔ Daryl's Win2K machine
  - » "follow the recipe"

“Dead” Analysis of Daryl’s Computer

10/26/2004

EMAIL:

Yahoo! Mail - darylpopper@yahoo.com - Mozilla Firefox

File Edit View Go Bookmarks Tools Help

file:///C:/Research/tutorials/acsac2006/Recipes/fakedfromignatius.htm

Getting Started Latest Headlines

Yahoo! Mail - nilesboudreaux@yahoo... | Yahoo! Mail - darylpopper@yah...

Yahoo! My Yahoo! Mail

Search: Web Search

YAHOO! MAIL

Welcome, darylpopper
[Sign Out, My Account]

Mail Home - Mail Tutorials - Help

THE ULTIMATE MEDIA RAZR from Verizon Wireless
U CAST COVERAGE & $15/MO UPAK SUBSCRIPTION REQ'D PER SONG CHARGES APPLY
With verizon wireless CAST SM MUSIC
Your phone is a music player.

Mail | Addresses | Calendar | Notepad

Mail For Mobile - Upgrades - Options

Check Mail | Compose | Search Mail | Search the Web

Blockbuster.com
Try it for FREE!

Folders [Add - Edit]
- Inbox (1)
- Draft
- Sent
- Bulk [Empty]
- Trash [Empty]

Search Shortcuts
- My Photos
- My Attachments

See your credit score: $0
Don't quit job 1 year degree
Bad Credit Refinance Rates
Degrees in as fast as 1 year

Previous | Next | Back to Messages

Delete | Reply | Forward | Spam | Move...

This message is not flagged. [Flag Message - Mark as Unread] Printable View

Date:	Mon, 23 Oct 2004 17:19:26 -0500
From:	I. Riley" <iriley@lickmyspoon.com> Add to Address Book Add Mobile Alert
To:	darylpopper@yahoo.com
Subject:	Recipe

Dear Daryl,

The recipe you sent is fabulous! We've already put the pate into production and initial taste tests are very positive. We're open to more business if you can provide it. Discretion is extremely important,
as you know. The patents your company holds on these recipes will be very hard to uphold (we're tweaking the recipes just in case, anyway) unless there's some evidence that the recipes were leaked.

Please delete this email after you read it.

Best,

Ignatius Q. Riley
Co-founder, LickMySpoon, LLC

Delete | Reply | Forward | Spam | Move...

Previous | Next | Back to Messages

Save Message Text | Full Headers

Check Mail | Compose | Search Mail | Search the Web

Done

"Live" Analysis of Daryl's Computer

using *Sysinternals* tools

pslist* > pslist.txt

pslist.txt - Notepad

File Edit Format View Help

```
PsList 1.26 - Process Information Lister
Copyright (C) 1999-2004 Mark Russinovich
Sysinternals - www.sysinternals.com

Process information for DARYLP:

Name                 Pid Pri Thd  Hnd   Priv        CPU Time    Elapsed Time
Idle                   0   0   1    0      0     0:22:17.062     0:22:15.078
System                 8   8  32  154     24     0:00:04.203     0:22:15.078
smss                 140  11   6   33   1092     0:00:00.093     0:22:15.078
csrss                164  13  10  256   1248     0:00:00.953     0:22:03.875
winlogon             184  13  17  356   5360     0:00:00.500     0:22:01.796
services             212   9  33  483   2324     0:00:02.875     0:21:59.375
lsass                224   9  15  257   1800     0:00:00.140     0:21:59.343
svchost              380   8   7  166    868     0:00:00.031     0:21:53.968
SPOOLSV              408   8  11   97   2012     0:00:00.031     0:21:53.562
svchost              440   8  21  254   1828     0:00:00.156     0:21:53.453
regsvc               480   8   2   30    236     0:00:00.015     0:21:50.703
mstask               516   8   7   93    544     0:00:00.015     0:21:50.093
explorer             716   8  18  292   3200     0:00:01.046     0:19:34.281
cmd                  764   8   1   24    288     0:00:00.093     0:18:11.984
pslist               920  13   2   79    608     0:00:00.015     0:00:02.156
```

Anything interesting?

*pslist → list active processes

pmdump* -list > pmdump-list.txt

pmdump-list.txt - Notepad

File Edit Format View Help

```
pmdump 1.2 - (c) 2002, Arne Vidstrom (arne.vidstrom@ntsecurity.nu)
           - http://ntsecurity.nu/toolbox/pmdump/

    0 - System idle process
    8 - System
  140 - smss.exe
  164 - csrss.exe
  184 - winlogon.exe
  212 - services.exe
  224 - lsass.exe
  380 - svchost.exe
  408 - SPOOLSV.EXE
  440 - svchost.exe
  480 - regsvc.exe
  516 - mstask.exe
  716 - explorer.exe
  764 - cmd.exe
  452 - pmdump.exe
```

Anything interesting?

*pmdump → process memory dump; dump out the content of a process' RAM allocation

handle* > handle.txt

handle.txt - Notepad

File Edit Format View Help

```
  2AC: File  (RW-)   C:\Documents and Settings\Administrator\Local Settings\Appl
  2B4: File  (---)   C:\Documents and Settings\Administrator\Local Settings\Appl
  2B8: Section       \BaseNamedObjects\HGFSMEMORY
  2EC: Section       \BaseNamedObjects\UrlZonesSM_Administrator
  310: Section       \BaseNamedObjects\mmGlobalPnpInfo
```

[snip]

```
  360: Section       \BaseNamedObjects\WDMAUD_Callbacks
  368: File  (RW-)   C:\Documents and Settings\Administrator\Local Settings\Temp
  3B8: Section       \BaseNamedObjects\RotHintTable
  3C4: Section       \BaseNamedObjects\C:_Documents and Settings_Administrator_L
  3D8: File  (R--)   C:\WINNT\system32\shell32.dll
  3DC: File  (R--)   C:\WINNT\system32\webvw.dll
  3E4: File  (RW-)   C:\Documents and Settings\Administrator\Cookies\index.dat
  3F4: File  (RW-)   C:\Documents and Settings\Administrator\Local Settings\Hist
  3FC: Section       \BaseNamedObjects\C:_Documents and Settings_Administrator_L
  40C: Section       \BaseNamedObjects\C:_Documents and Settings_Administrator_C
  450: File  (RWD)   C:\intellectual
------------------------------------------------------------------------------
cmd.exe pid: 764 DARYLP\Administrator
------------------------------------------------------------------------------
<Non-existant Process> pid: 704 DARYLP\Administrator
   18: File  (RW-)   C:\taxes
------------------------------------------------------------------------------
handle.exe pid: 488 DARYLP\Administrator
```

!?

*handle → show open handles (e.g., files)

```
dd if=\\.\PhysicalMemory of=PhysicalMemory.dd
ptfinder_w2k.pl   PhysicalMemory.dd > ptfinder.txt
```

ptfinder.txt

No.	Type	PID	TID	Time created	Time exited	Offset	PDB	Remarks
1	Proc	0				0x0046bb60	0x00030000	Idle
2	Thrd	0	0			0x0046bdf0		
3	Thrd	0	0			0x00e62df0		
4	Proc	856		2004-10-26 14:20:20		0x015715e0	0x0d820000	dd.exe
5	Thrd	380	600			0x015718a0		
6	Thrd	904	612		2004-10-26 14:19:50	0x01571b20		
7	Thrd	224	520			0x01571da0		
8	Thrd	8	904			0x01574480		
9	Thrd	224	876			0x01574700		
10	Thrd	8	928			0x01575da0		
11	Thrd	440	916			0x01590da0		
12	Thrd	408	924			0x015912e0		
13	Thrd	440	888			0x01591da0		
14	Thrd	716	720			0x01596540		
15	Thrd	440	908		2004-10-26 14:10:26	0x015967c0		
16	Thrd	784	752		2004-10-26 14:19:39	0x01598020		
17	Thrd	716	756			0x01599d20		
18	Thrd	440	492			0x0159b0a0		
19	Thrd	716	112		2004-10-26 14:20:39	0x0159b420		
20	Thrd	440	868		2004-10-26 14:10:26	0x0159bbc0		
21	Thrd	380	444			0x0159dda0		
22	Thrd	440	660		2004-10-26 14:11:09	0x0159e020		
23	Proc	704		2004-10-22 11:50:36		0x015ac900	0x07bcc000	turbotax97.exe

*dd → byte level copy

** ptfinder → extract running/exited processes

`turbotax97.exe` EXAMINATION

- Actually, a ***keylogger***
- Logs stored in **c:\taxes**
  - **log.txt** file in directory

NETWORK VIEW

TCPView - Sysinternals: www.sysinternals.com

File Options Process View Help

Proce...	Protocol	Local Address	Remote Address	State
lsass.exe:224	UDP	darylp:isakmp	*:*	
mstask.exe:516	TCP	DARYLP:1025	DARYLP:0	LISTENING
services.exe:212	UDP	DARYLP:1026	*:*	
svchost.exe:380	TCP	DARYLP:epmap	DARYLP:0	LISTENING
svchost.exe:380	UDP	DARYLP:epmap	*:*	
System:8	TCP	DARYLP:microsoft-ds	DARYLP:0	LISTENING
System:8	TCP	DARYLP:1028	DARYLP:0	LISTENING
System:8	TCP	DARYLP:1027	localhost:microsoft-ds	TIME_WAIT
System:8	TCP	darylp:netbios-ssn	DARYLP:0	LISTENING
System:8	TCP	darylp:netbios-ssn	sparelaptop3:1029	ESTABLISHED
System:8	TCP	darylp:1031	sparelaptop3:netbios-ssn	TIME_WAIT
System:8	UDP	DARYLP:microsoft-ds	*:*	
System:8	UDP	darylp:netbios-ns	*:*	
System:8	UDP	darylp:netbios-dgm	*:*	

Why is the computer talking to a spare laptop?

- We don't know what the traffic is
  - but it **does** look suspicious
- Immediately institute network logging on corporate network
  - special attention paid to traffic involving **sparelaptop3**
- Also, begin investigation of `sparelaptop3`, which sits in the company print room

Network analysis w/ Wireshark

10/24/2004

Files Xfer from Daryl → Spare Found in Network Traces

- **bobotie-notes.txt.bfe**
  - » Directory info enumerated at packet 2029-2030.
  - » 662 bytes (entire file) transferred in packet 2202.

- **pate.txt** (un-encrypted)
  - » choose any packet in stream and follow TCP stream, e.g., packet 3360.
  - » data transfer is actually at packet 3479.

- **pistachio-macaroons.txt.bfe**
  - » 246 bytes (entire file) transferred in packet 4693.

Follow TCP Stream

Stream Content

```
\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p... .
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.............p.SMB.....p...0.....
\.i.n.t.e.l.l.e.c.t.u.a.l.......SMB.....p...0.*.....
@....`w.o.......[......Nm......Nm.....a.l.....n.d.o.w.s. .2........2...H.SMB2.....p...@.....D.....e.
@.....<.SMB2.....p...@.
........8.....<.....H.SMB2.....p...P.......(.....D.....e.
@.....d.SMB2.....p...P.
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....).SMB.....`..
@.........#.SMB.....`.....l.SMB2.....p...p..(...
..@.........(.D........+......V........\.i.n.t.e.l.l.e.c.t.u.a.l.......SMB2.....p...p.

.|...
.8...|.D.....`w.o.....Nm.....Nm.....Nm.....I.N.T.E.L.L.~.1........i.n.t.e.l.l.e.c.t.u.a.l.........f.SMB
2.....p.....".....(.....".D........%.........\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p.....
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....f.SMB2.....p.....".....(.....".D........%.........
\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p.....
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....l.SMB2.....p.....(...
..@.........(.D........+......V........\.i.n.t.e.l.l.e.c.t.u.a.l.......SMB2.....p.....

.|...
.8...|.D.....`w.o.....Nm.....Nm.....Nm.....I.N.T.E.L.L.~.1........i.n.t.e.l.l.e.c.t.u.a.l.........f.SMB
2.....p.....".....(.....".D........%.........\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p.....
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....f.SMB2.....p.....".....(.....".D........%.........
\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p.....
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....p.SMB.....p.....
\.i.n.t.e.l.l.e.c.t.u.a.l.......SMB.....p.....*......@....`w.o.......
[......Nm......Nm.....a.l.....H.SMB2.....p.....D.....e..@.....<.SMB
2.....p.....
........8.....<.....H.SMB2.....p.....(.....D.....e..@.....d.SMB2.....p.....
..(.....8...(.<.....-......`w.o.......
[......Nm......Nm.....).SMB.....@.........#.SMB.....l.SMB2.....p.....(...
..@.........(.D........+......V........\.i.n.t.e.l.l.e.c.t.u.a.l.......SMB2.....p.....

.|...
.8...|.D.....`w.o.....Nm.....Nm.....Nm.....I.N.T.E.L.L.~.1........i.n.t.e.l.l.e.c.t.u.a.l.........f.SMB
2.....p... ."....(.....".D........%.........\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p... .
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....f.SMB2.....p...0.".....(.....".D........%.........
\.i.n.t.e.l.l.e.c.t.u.a.l.....d.SMB2.....p...0.
..(.....8...(.<.....-......`w.o.......[......Nm......Nm.....SMB2.....p...@..N.....(.....N.D........Q..A.....\.i.n.t.e.l.l.e.c.t.u.a.l.
\.b.o.b.o.t.i.e.-.n.o.t.e.s...t.x.t...b.f.e.....d.SMB2.....p...@.
..(.....8...(.<.....-......`w.o....p.......`w.o....`w.o.... .....SMB2.....p...P..N.....(.....N.D........Q..A.....\.i.n.t.e.l.l.e.c.t.u.a.l.
\.b.o.b.o.t.i.e.-.n.o.t.e.s...t.x.t...b.f.e.....d.SMB2.....p...P.
..(.....8...(.<.....-......`w.o....p.......`w.o....`w.o.... .....SMB.....p...`.......F.....I..
\.i.n.t.e.l.l.e.c.t.u.a.l.
\.b.o.b.o.t.i.e.-.n.o.t.e.s...t.x.t...b.f.e.....SMB.....p...`.*......@....`w.o....p.......`w.o....`w.o.... .....a.l.
\.b.o.b.o.t.i.e.-.........s...H.SMB2.....p...p.....D.....e..@.....b.SMB2.....p...p.
..&.....8...&.<.....+.....:.:.$.D.A.T.A....H.SMB2.....p.....
(.....D.....e..@.....d.SMB2.....p.....
..(.....8...
(.<.....-......`w.o....p.......`w.o....`w.o.... .....;.SMB.....@.....T.SMB.....
<.....T.....:.....H\..3.....SMB.....p.....\.i.n.t.e.l.l.e.c.t.u.a.l.
\.b.o.b.o.t.i.e.-.n.o.t.e.s...t.x.t...b.f.e.:...R.a.e.c.2.5.p.h.4.s.u.d.b.f.0.h.A.a.q.5.e.h.w.3.N.f.:.
$.D.A.T.A......#.SMB.4.....p.....).SMB.....@.....#.SMB.....SMB.....
p.....F.....I..\.i.n.t.e.l.l.e.c.t.u.a.l.\.b.o.b.o.t.i.e.-.n.o.t.e.s...t.x.t...b.f.e.....SMB.....p.....*.....
@....`w.o.....".
^....`w.o....`w.o.... .....a.l.\.b.o.b.o.t.i.e.-.....s...H.SMB2.....p.....D.....e.
@.....b.SMB2.....p.....
..&.....8...&.<.....+.....:.:.$.D.A.T.A....H.SMB2.....p.....(.....D.....e.
```

Find | Save As | Print | Entire conversation (68221 bytes) | ASCII | EBCDIC | Hex Dump | C Arrays | Raw

Help | Filter Out This Stream | Close

network-trace.pcap - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: tcp.stream eq 2 Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
2195	237.875855	192.168.229.130	192.168.229.129	SMB	Trans2 Request, QUERY_FILE_INFO, FID: 0x8000, Query File Stream Info
2196	237.876776	192.168.229.129	192.168.229.130	SMB	Trans2 Response, FID: 0x8000, QUERY_FILE_INFO
2197	237.876778	192.168.229.130	192.168.229.129	SMB	Trans2 Request, QUERY_FILE_INFO, FID: 0x8000, Query File Basic Info
2198	237.879748	192.168.229.129	192.168.229.130	SMB	Trans2 Response, FID: 0x8000, QUERY_FILE_INFO
2199	237.884087	192.168.229.130	192.168.229.129	SMB	Trans2 Request, QUERY_FILE_INFO, FID: 0x8000, Query File Standard Info
2200	237.884098	192.168.229.129	192.168.229.130	SMB	Trans2 Response, FID: 0x8000, QUERY_FILE_INFO
2201	237.884813	192.168.229.130	192.168.229.129	SMB	Read AndX Request, FID: 0x8000, 4096 bytes at offset 0
2202	237.885196	192.168.229.129	192.168.229.130	SMB	Read AndX Response, FID: 0x8000, 662 bytes
2203	237.885343	192.168.229.130	192.168.229.129	SMB	Read AndX Request, FID: 0x8000, 3434 bytes at offset 662
2204	237.885421	192.168.229.129	192.168.229.130	SMB	Read AndX Response, FID: 0x8000, 0 bytes
2205	237.886348	192.168.229.130	192.168.229.129	SMB	Close Request, FID: 0x8000
2206	237.886366	192.168.229.129	192.168.229.130	SMB	Close Response, FID: 0x8000
2207	238.011546	192.168.229.130	192.168.229.129	TCP	solid-mux > netbios-ssn [ACK] Seq=13239 Ack=14487 Win=63118 Len=0
2208	238.045095	192.168.229.130	192.168.229.129	SMB	Trans2 Request, QUERY_PATH_INFO, Query File Basic Info, Path: \intellectual
2209	238.045239	192.168.229.129	192.168.229.130	SMB	Trans2 Response, QUERY_PATH_INFO
2210	238.045372	192.168.229.130	192.168.229.129	SMB	Trans2 Request, QUERY_PATH_INFO, Query File Basic Info, Path: \intellectual
2211	238.045439	192.168.229.129	192.168.229.130	SMB	Trans2 Response, QUERY_PATH_INFO
2212	238.046325	192.168.229.130	192.168.229.129	SMB	Trans2 Request, QUERY_PATH_INFO, Query File Basic Info, Path: \intellectual

```
⊞ Ethernet II, Src: VMware_28:cc:68 (00:0c:29:28:cc:68), Dst: VMware_73:7a:0b (00:0c:29:73:7a:0b)
⊞ Internet Protocol, Src: 192.168.229.129 (192.168.229.129), Dst: 192.168.229.130 (192.168.229.130)
⊞ Transmission Control Protocol, Src Port: netbios-ssn (139), Dst Port: solid-mux (1029), Seq: 13659, Ack: 13131, Len: 726
⊞ NetBIOS Session Service
⊟ SMB (Server Message Block Protocol)
  ⊞ SMB Header
  ⊟ Read AndX Response (0x2e)
    ⊞ [FID: 0x8000 (\intellectual\bobotie-notes.txt.bfe)]
      Word Count (WCT): 12
      AndXCommand: No further commands (0xff)
      Reserved: 00
      AndXOffset: 0
      [File Offset: 0]
      [File RW Length: 4096]
      Remaining: 65535
      Data Compaction Mode: 0
      Reserved: 0000
      Data Length Low: 662
      Data Offset: 60
      Data Length High (multiply with 64K): 0
      Reserved: 000000000000
      Byte Count (BCC): 663
      Padding: 01
      File Data: 5401CCC392B0903A81B110C715F00F485CC7B133D7EA92D6...
```

```
0070  00 00 00 97 02 01 54 01  cc c3 92 b0 90 3a 81 b1   ......T. .....:..
0080  10 c7 15 f0 0f 48 5c c7  b1 33 d7 ea 92 d6 4f 0f   .....H\. .3....O.
0090  d2 3e a9 59 e1 67 59 42  98 36 f8 81 53 d6 da dd   .>.Y.gYB .6..S...
00a0  f5 9f ab 03 f6 1b e3 2b  f1 45 fb ba 2c b2 e5 b6   .......+ .E..,...
00b0  bb 4c e3 89 f2 4b 9c 38  5c 13 0b 69 d9 70 c6 f7   .L...K.8 \..i.p..
00c0  [illegible]
```

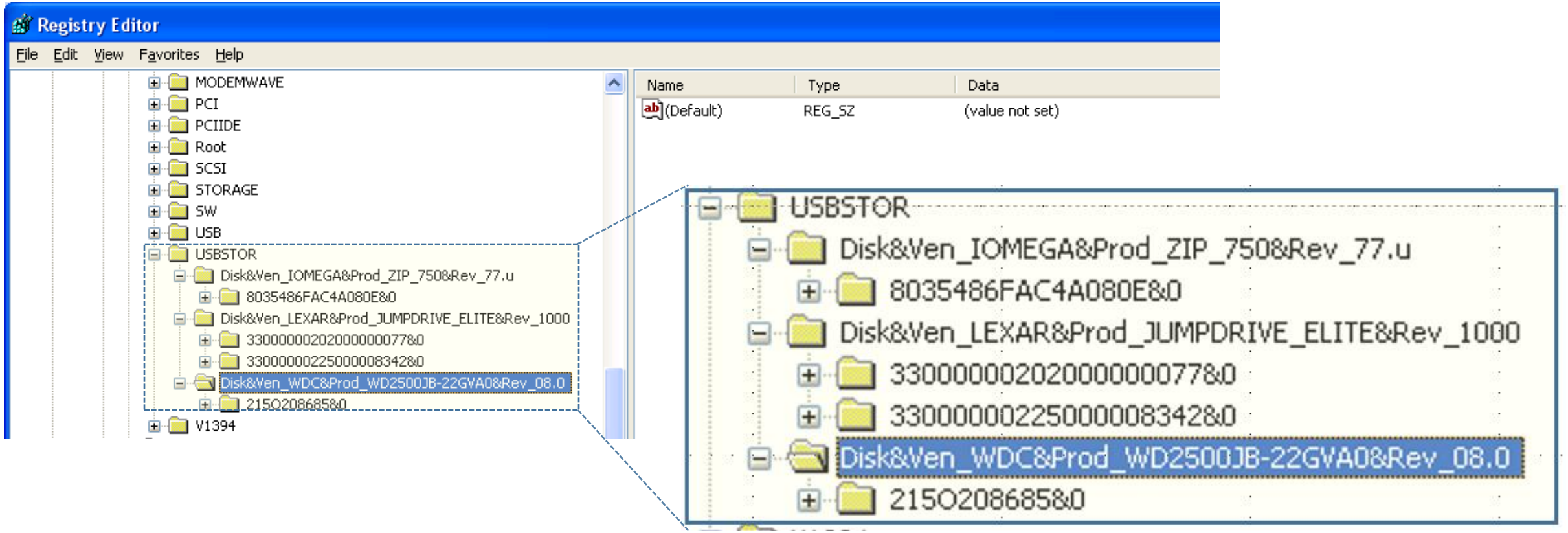
Data read/written to the file (smb.file_data), 66... | Packets: 5195 Displayed: 580 Marked: 0 | Profile: Default

C:\WINDOWS\system32\cmd.exe - less bototie-notes.txt

```
************* INTELLECTUAL PROPERTY OF TURBOCHEF, LLC *************
********UNAUTHORIZED ACCESS OR DUPLICATION IS PROHIBITED***********
Revision: 1.53
Custodian: darylp

Bototie varation w/ Bulghur

1 cups bread crumbs
1.5 cups milk
2 cups onions, finely chopped
1 apple, finely chopped
1 clove garlic, minced
3 T butter
1/3 cup dried blueberries or cherries
1/4 cup slivered almonds
2 T curry powder
1 t sugar
3 large eggs
1.5lbs lamb
1/2 cup bulghur
2 T lemon juice
1/2 t lemon zest
1/2 t black pepper


Soak bulghur in hot water and and then press out water.
Soak bread crumbs in milk. Strain and reserve milk.

Saute onions, garlic, apples, and 1/4t salt until onions are clear.
Add dried fruit, almonds, curry powder, and sugar, and cook for 1
minute.  Remove from heat.

Beat 1 egg and mix in lamb, onion mixture, strained bread crumbs, bulghur, 1.25t salt,
and black pepper.

Bake at 325 for 30 mins.

Whisk milk from bread crumbs, 2 eggs, and 1/4 t salt.

Pour custard over meatloaf and return to oven and bake until cooked through.


bototie-notes.txt (END)
```

USB INSERTION HISTORY: LOANER LAPTOP

HKEY_LOCAL_MACHINE/SYSTEM/CURRENTCONTROLSET/ENUM/USBSTOR



USB storage belonging to Niles Boudreaux, another TurboChef employee!

USB insertion history: Niles Boudreaux's machine

HKEY_LOCAL_MACHINE/SYSTEM/CurrentControlSet/Enum/USBSTOR

Registry Editor

File Edit View Favorites Help

- Vid_050d&Pid_0237
  - 5&697b0788&0&4
- Vid_050d&Pid_0248
  - A00000010F67
    - Control
    - Device Parameters
    - LogConf
- Vid_05dc&Pid_a400
  - 33000000202000000077
    - Control
    - Device Parameters
    - LogConf
  - 33000000225000008342
    - Device Parameters
    - LogConf
- Vid_0644&Pid_0200
- Vid_0781&Pid_5200
- Vid_0c0b&Pid_27e7
- Vid_0d49&Pid_3210
- Vid_1058&Pid_0900
- Vid_413c&Pid_1003
- Vid_413c&Pid_2010
- Vid_413c&Pid_2010&MI_00
- Vid_413c&Pid_2010&MI_01
- USBSTOR
  - Disk&Ven_BELKIN&Prod_USB_2_HS-CF&Rev_1.95
  - Disk&Ven_BELKIN&Prod_USB_2_HS-MS&Rev_1.95
  - Disk&Ven_BELKIN&Prod_USB_2_HS-SD/MMC&Rev_1.95
  - Disk&Ven_BELKIN&Prod_USB_2_HS-SM&Rev_1.95
  - Disk&Ven_LEXAR&Prod_JUMPDRIVE_ELITE&Rev_1000
    - 33000000202000000077&0
      - Control
      - Device Parameters

Name	Type	Data
(Default)	REG_SZ	(value not set)
Capabilities	REG_DWORD	0x00000010 (16)
Class	REG_SZ	DiskDrive
ClassGUID	REG_SZ	{4D36E967-E325-11CE-BFC1-08002BE10318}
CompatibleIDs	REG_MULTI_SZ	USBSTOR\Disk USBSTOR\RAW
ConfigFlags	REG_DWORD	0x00000000 (0)
DeviceDesc	REG_SZ	Disk drive
Driver	REG_SZ	{4D36E967-E325-11CE-BFC1-08002BE10318}\0018
FriendlyName	REG_SZ	LEXAR JUMPDRIVE ELITE USB Device
HardwareID	REG_MULTI_SZ	USBSTOR\DiskLEXAR___JUMPDRIVE_ELITE_1000 USBSTOR\DiskLEXAR...
Mfg	REG_SZ	(Standard disk drives)
ParentIdPrefix	REG_SZ	7&195ac47a&0
Service	REG_SZ	disk
UINumber	REG_DWORD	0x00000000 (0)

Matching records
➔ same USB stick used on both machines!

- Disk&Ven_LEXAR&Prod_JUMPDRIVE_ELITE&Rev_10
  - 33000000202000000077&0
    - Control

On Niles Boudreaux's Computer

- Copies of encrypted and unencrypted recipes (deleted)
- Original email from Ignatius Q. Riley (deleted)

Actual Email From Ignatius (from analysis of N. Boudreaux's machine)

Yahoo! Mail - nilesboudreaux@yahoo.com - Mozilla Firefox

file:///C:/Research/tutorials/acsac2006/Recipes/letterfromignatius.htm

Welcome, **nilesboudreaux**
[Sign Out, My Account]

Mail Home - Mail Tutorials - Help

THE ULTIMATE MEDIA RAZR from Verizon Wireless
Only $49.99 After $50 Instant online discount WITH NEW 2 YR ACTIVATION
BUY NOW
verizon wireless
MOTOROLA U CAST RAZR U3M HEADSET SOLD SEPARATELY

Mail | Addresses | Calendar | Notepad

Mail For Mobile - Upgrades - Options

Check Mail | Compose | Search Mail | Search the Web

Blockbuster.com
Try it for **FREE!**

Folders [Add - Edit]
- Inbox (1)
- Draft
- Sent
- Bulk [Empty]
- Trash [Empty]

Search Shortcuts
- My Photos
- My Attachments

See your credit score: $0
Don't quit job 1 year degree
Bad Credit Refinance Rates
Degrees in as fast as 1 year

Previous | Next | Back to Messages

Delete | Reply | Forward | Spam | Move...

This message is not flagged. [Flag Message - Mark as Unread] Printable View

Date:	Mon, 23 Oct 2004 17:19:26 -0500
From:	I. Riley" <iriley@lickmyspoon.com> Add to Address Book Add Mobile Alert
To:	nilesboudreaux@yahoo.com
Subject:	Recipe

Dear Niles,

The recipe you sent is fabulous! We've already put the pate into production and initial taste tests are very positive. We're open to more business if you can provide it. Discretion is extremely important,
as you know. The patents your company holds on these recipes will be very hard to uphold (we're tweaking the recipes just in case, anyway) unless there's some evidence that the recipes were leaked.

Please delete this email after you read it.

Best,

Ignatius Q. Riley
Co-founder, LickMySpoon, LLC

Delete | Reply | Forward | Spam | Move...

Previous | Next | Back to Messages

Save Message Text | Full Headers

Check Mail | Compose | Search Mail | Search the Web

Done

A Closer Look at Email on Daryl Popper's Computer

Yahoo! Mail - darylpopper@yahoo.com - Mozilla Firefox

File Edit View Go Bookmarks Tools Help

file:///C:/Research/tutorials/acsac2006/Recipes/fakedfromignatius.htm

Getting Started Latest Headlines

Yahoo! Mail - nilesboudreaux@yahoo... | Yahoo! Mail - darylpopper@yah...

Yahoo! My Yahoo! Mail

Search: Web Search

YAHOO! MAIL

Welcome, darylpopper
[Sign Out, My Account]

Mail Home - Mail Tutorials - Help

THE ULTIMATE MEDIA RAZR from Verizon Wireless
U CAST COVERAGE & $15/MO UPAK SUBSCRIPTION REQ'D PER SONG CHARGES APPLY
With verizon wireless CAST SM MUSIC
Your phone is a music player.

Mail | Addresses | Calendar | Notepad

Mail For Mobile - Upgrades - Options

Check Mail | Compose | Search Mail | Search the Web

Blockbuster.com
Try it for FREE!

Folders [Add - Edit]
- Inbox (1)
- Draft
- Sent
- Bulk [Empty]
- Trash [Empty]

Search Shortcuts
- My Photos
- My Attachments

See your credit score: $0
Don't quit job 1 year degree
Bad Credit Refinance Rates
Degrees in as fast as 1 year

Previous | Next | Back to Messages

Delete | Reply | Forward | Spam | Move...

This message is not flagged. [Flag Message - Mark as Unread] Printable View

Date:	Mon, 23 Oct 2004 17:19:26 -0500
From:	I. Riley" <iriley@lickmyspoon.com> Add to Address Book Add Mobile Alert
To:	darylpopper@yahoo.com
Subject:	Recipe

Dear Daryl,

The recipe you sent is fabulous! We've already put the pate into production and initial taste tests are very positive. We're open to more business if you can provide it. Discretion is extremely important,
as you know. The patents your company holds on these recipes will be very hard to uphold (we're tweaking the recipes just in case, anyway) unless there's some evidence that the recipes were leaked.

Please delete this email after you read it.

Best,

Ignatius Q. Riley
Co-founder, LickMySpoon, LLC

Delete | Reply | Forward | Spam | Move...

Previous | Next | Back to Messages Save Message Text | Full Headers

Check Mail | Compose | Search Mail | Search the Web

Done

A Closer Look at Email on Daryl Popper's Computer (**HTML SOURCE**)

The Real Story

1. Niles unhappy, contacts Lick-My-Spoon
2. Niles installs key logger on Daryl's machine while she's at lunch, and enables guest account and file sharing
3. Key logger records Daryl' s encryption passwords
4. Niles digs up unused, spare laptop
5. Niles gets keystroke log remotely using spare laptop / file sharing
6. Niles copies some recipes to spare laptop using file sharing
7. Niles copies stolen recipes from spare laptop to USB disk
8. Niles takes USB disk to his computer, decrypts, sends recipe to Ignatius
9. Ignatius sends reply to Niles
10. Niles edits Ignatius' reply, making it look like it was sent to Daryl (oops)
11. Niles deposits forged reply on Daryl's machine
12. Someone notices Turbo-Chef's recipe at Lick-My-Spoon, alerts CEO, investigation begins
13. Niles continues periodically to copy recipes from Daryl's machine to spare laptop
14. (These transfers were captured during network logging)
15. Facing mounting evidence, Niles confesses